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Citation for the original published paper (version of record):

Maharjan, R., Vessby, J., Kuai, L., Ormarsson, S. (2026)
A numerical study of multi-story light-frame timber modular towers with rubber based inter-modular connections using substructure method
Engineering structures, 369(2): 123831
https://doi.org/10.1016/j.engstruct.2026.123831

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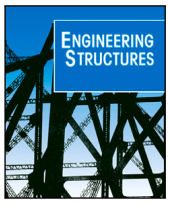
Engineering Structures 369 (2026) 123831



Contents lists available at ScienceDirect

Engineering Structures

journal homepage: www.elsevier.com/locate/engstruct



A numerical study of multi-story light-frame timber modular towers with rubber based inter-modular connections using substructure method

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ARTICLE INFO

Keywords:
Timber modules
Adaptive model
Substructure
Inter-module connections

ABSTRACT

With the growing interest in timber building technologies and the pursuit of efficient modular construction, light-frame timber modules are gaining increasing popularity across the globe. Most residential buildings using such modules in Nordic countries are built up to a maximum six stories and are typically designed for the racking strength using the simplified analysis of wall diaphragms prescribed in Eurocode 5. Recent studies in light-frame timber shear walls present detailed finite element (FE) models that are better than simplified analysis. However, for modular systems, apart from the shear walls, the influence of transverse walls, floor diaphragms and inter-modular connections plays a crucial role in accurately predicting the global response of multi-story buildings. Although creating a detailed FE model of multiple modules can be computationally demanding, excessive simplification may lead to loss of essential structural behavior. To address this, the present study proposes an adaptive modeling strategy that offers a balance between accuracy and computational efficiency. The approach combines a non-linear standard FE-formulation for a representative module with a linear substructure formulation for stacked modules, which may have identical or varying designs. The substructure modules are connected to nonlinear inter-modular connections via nodal points where the degrees of freedom are retained. Consequently, the limitation of the substructure formulation, namely, the continuation of linear behavior beyond plastic deformation of the module, is confined to the substructure models meanwhile, the nonlinear effects are captured in the critical bottom module and at the inter-modular connections. A tower is analyzed with an adaptive model of a typical asymmetric module to evaluate both the global racking stiffness and racking strength, as well as the local sheathing-to-framing connection and axial compression in studs of the most heavily loaded bottom module. Story drift of only the bottom module of the six story tower is found to be within the recommended serviceability limit. For the modules above, around 40% contribution to the total horizontal displacement was due to the rigid body rotation, or rocking, of the module while using friction based inter-modular connections. An additional 35% contribution was from shear displacement of the modules while the remaining 25% was caused by slip displacement. Stiffer design of inter-modular connections is crucial and therefore, is recommended in design practice to reduce the story drift, thus upgrading the design to reach the recommended serviceability limit criteria.

1. Introduction

In recent decades, mid-rise residential buildings made from prefabricated timber modules have gained growing interest, especially in Nordic countries. These buildings are attractive because of their low environmental impact, better working conditions due to the indoor manufacturing process during construction and fast assembly on-site [1]. The fully volumetric modular system units are designed according to Eurocode 5 (EC5) [2], produced in factory under climate control and transported to the construction site with internal and external finishes as well as acoustic and thermal insulation.

There are two main ways to design buildings using prefabricated modules: frame-supported and self-supporting systems. In frame-supported buildings, modules are placed inside a load-bearing frame, which carries most of the external loads [3]. This significantly reduces

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https://doi.org/10.1016/j.engstruct.2026.123831
Received 23 January 2026; Received in revised form 5 August 2026; Accepted 18 September 2026
Available online 1 October 2026
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the forces acting through the modules themselves. In self-supporting buildings, on the other hand, both vertical and horizontal loads are transferred through the modules via structural elements like floors, roof diaphragms and walls. Under lateral loading, some of these walls act as shear walls and are subjected to in-plane shear action.

Conventional light-framed timber shear walls or wall diaphragms with rails, studs and sheathing panels attached by mechanical fasteners [4] have been widely studied and simplified design methods for their racking stiffness and racking strength are available, for example, in EC 5 [2]. However, when the shear walls are part of a prefabricated module, their interaction with floor diaphragms and transverse walls affects their boundary conditions and behavior, which change the precondition for the analysis significantly. The shear wall interaction with other components of the module is a part of the system effect [5]. There may also be large differences in the analysis due to the module geometry, number of sheathing layers, behavior of intra-modular connections linking shear walls to the transverse walls and floor diaphragms, and the design to resist uplift forces.

In addition, the inter-modular connections linking two modules vertically or horizontally are important for load transfer as well as obtaining desired vibro-acoustic properties [6]. Generally, those connections are metal plates with screws or mechanical slots for other types of modular construction [7] but in the construction practice adopted by timber module building companies in the Nordic countries, friction based inter-modular connections (elastomer bricks) are commonly used because of their superior vibro-acoustic performance reducing sound and vibrations between apartments within the same building.

The behavior of the unit volumetric light-frame timber module have been studied in a full-scale experimental test series [8] and a numerical model to verify the experimental results has previously been developed [9]. Similar to the light-frame shear wall, the system effect in an assembly of several modules, in vertical and horizontal directions in a modular building, depends on vertical and horizontal inter-modular connections and intra-modular connections. Research studies of timber modular buildings are very limited to the authors' knowledge [10,11] but the system effect and importance of the inter-modular connections are studied in timber frame buildings [12], cross-laminated timber (CLT) structures [13,14] and in steel modular buildings e.g. [5,15].

Given all of these preconditions, using finite element (FE) analysis is a powerful way to better understand how these buildings behave while loaded. Full-building simulations make it possible to analyze how forces are distributed within and between modules, and can be used to identify the most critical failure modes under racking loads. However, modeling an entire building in detail can be time consuming and computationally costly, especially if the goal is to capture both global behavior, overall stiffness and strength, and local effects such as forces in each individual sheathing-to-framing connection. To manage this, simplifications can be made to reduce the computational cost while maintaining sufficient accuracy. For example, substructuring techniques can be used to simplify how individual modules are modeled. This involves selecting only a limited number of degrees of freedom (DoFs) to keep in the analysis, known as "retained" DoFs, while the rest are treated as secondary DoFs in which results are not explicitly calculated.

One drawback of this method is that it excludes nonlinear behavior within the module, which might lead to less accurate results. However, this can be addressed through a pre-analysis to check whether the individual module with standard FE-formulation remains linear under expected load levels. If significant nonlinear behavior is likely, the simplified "substructure" model or superelement can be replaced with a "standard" FE model that includes explicit calculations of all DoFs and any nonlinear material behavior. This paper explores how to accurately simulate the behavior of self-supporting mid-rise timber buildings made from prefabricated modules under combined gravity and lateral loading, using a suitable balance between computational efficiency and precision.

1.1. Literature review

Considerable amount of research work has been put into the broad topic of racking stiffness and racking strength during the past few decades. Relevant research covers topics ranging from the behavior of single fasteners connecting sheathing panels to framing members, to assemblies of wall panels including openings, and to some extent, variation in boundary conditions and vertical loading [16–18]. There are thorough literature reviews addressing sheathed wood-framed shear walls available [4,19,20] and the reader is referred to that or other works for a more general background on the topic. The objective of the work carried-out has mainly been to develop methods for predicting the racking stiffness and racking strength of relatively simple shear walls. Examples of analytical expressions developed for predicting at least one of the two can be found, e.g. [4,21–25]. These types of derived equations play an important role when engineers and researchers seek methods to design simpler shear wall structures. However, there are certain important shortcomings of these models which may lead to poor results, and some of the more important ones are:

1. The effect of uncertain distribution of vertical load.
2. Boundary conditions are typically severely simplified.
3. The nonlinear behavior of any fastener is omitted, and tough simplifications are made regarding their behavior, which is often assumed to be linear elastic, ideally plastic.
4. The geometry of an entire wall assembly is often complex and thus is not easily included in the models.

These and other shortcomings of analytical models have led to the development of more or less sophisticated FE models that also cover the range spanning from the single fastener [26–28] to the wall panel assemblies [29–33]. An important role of the FE models developed is to gain insights into the actual behavior of the entire assemblies. For instance, simulations of wall panel assemblies can help to understand the occurrence of crack initiations in sheathing panels [34] and account for the contribution of different boundary conditions and geometry, varying vertical load, and geometrical or material nonlinearities. Thus, it is of interest to extend the models even further and to develop FE models of entire prefabricated modules. Such modules have been studied lately, e.g. [9,35] with respect to their structural behavior during loading, and they have also been validated by experimental tests [8]. A similar FE model of prefabricated timber modules were previously developed [3] in which dynamic behavior of a detailed model was compared to simplified shear frames or trussed frames and calibrated against the test results. The detailed model was too complicated and computationally costly to incorporate in the global analysis whereas the simplified models needs to be calibrated to have correct dynamic behavior.

FE modeling of inter-modular connections for steel modules have previously been performed e.g. [36–39] and the results show the significance of those connections in global behavior of the modular buildings. A novel 3D printed inter-modular (sliding and stacking) connection system for CLT modules was explored [40] for efficient and flexible assembly in volumetric construction. Dynamic substructure simulation models for elastomer bricks have been suggested [41] to evaluate the performance of reduction methods by dividing the structure at the interfaces of elastomers and surrounding parts. A study on the dynamic properties of rubbers was conducted [42] in which a new visco-plastic model for rubber materials used in engineering applications was purposed. A FE simulation was performed to characterize elastomers and determine their material properties in terms of noise and vibration transmission [43].

The seismic modeling of mid-rise light-frame timber buildings using a simplified coupled shear-bending model [44,45] compared the seismic response against full-scale experiments and predicted building displacements, inter-story drifts and anchor tie down system. A simple link-frame model that models shear walls with frame elements and links

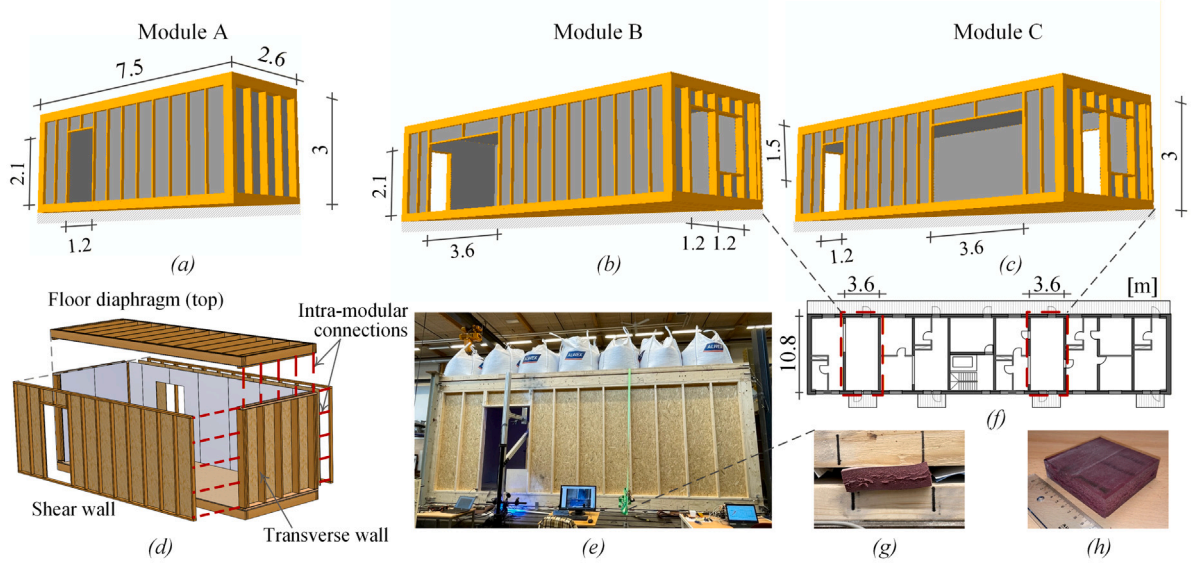


Fig. 1. Geometry of (*a*) module A, (*b*) typical module B, (*c*) typical module C, (*d*) elements of the module A, (*e*) experimental module A, (*f*) extracts of typical modules from a modular building plan, (*g*) deformed elastomer brick e_b during the experiment and (*h*) undeformed elastomer brick.

is suggested in [12] to analyze wind and seismic performance of timber frame multi-story buildings. The model can capture building system effects or the interaction of shear walls with other assemblies but cannot provide local behaviors like sheathing-to-framing forces. Analysis of intra-modular connection in CLT panels with dowel-type fasteners and mechanical anchors show the enhanced lateral stiffness, capacity and deformation [46]. These numerical studies on intra-modular and inter-modular connections illustrate their importance in modular buildings, however too much simplification in modeling modules may lead to inaccurate estimate of the multi-story buildings and therefore a detailed model capable of simulating local behavior of the modules is deemed necessary.

1.2. Present study

Studies of planar light-frame timber structures in multi-story building overshadows volume based timber modular buildings and detailed research work on modeling of the modules are hard to find in previous research. In addition, a few studies on friction based inter-modular connections have been performed together with modular units. This paper aims to scrutinize the behavior of multi-story timber modular building including inter-modular connections by analyzing multiple vertically stacked timber modules interacting with each other via rubber based elastomer bricks. The analysis is based on a previously developed FE model [9,47], which is here extended to include interactions between multi-story modules through their inter-modular connections. To make the analysis more efficient and reduce computational costs, multi-point kinematic constraints and substructure techniques are used. The idea of using substructure is reduction of DoFs [48] and the technique has been implemented in several FE software tools, including Abaqus [49]. Such substructures can be coupled and used in static, and even in dynamic analyses [50].

The aim is also to understand how forces are distributed through an entire "tower" of stacked modules, and to identify critical structural aspects of self-supporting modular buildings subjected simultaneously to gravity loads and static horizontally oriented wind loads. The analysis accounts for all structural elements, including each individual fastener. The racking stiffness and racking strength for the tower are analyzed using code based design loads and the obtained serviceability limit state displacements are compared to the recommended limits. In the study a detailed FE analysis of the most heavily loaded module in the bottom story is combined with substructure analysis of the other stacked modules. This adaptive model is a tool for performance evaluation of modular buildings currently in practice and designing mid-rise modular buildings and the substructure method for static analysis is used in order to reduce the computational cost. Despite this, there are some limitations including:

1. Verification of the FE model is based on a single module rather than a stack of modules.
2. Displacements caused locally by effects of compression perpendicular to the grain in horizontal members is excluded.
3. A linear horizontal force vs vertical compression P_x-v_y relation for the coupled spring of inter-modular connections in vertical direction is assumed and
4. side-wise interaction between modules (inter-modular connections in horizontal direction) are omitted.

2. Methodology

2.1. Geometry of modules

In this study an "experimental" module and two types of "typical" modules are included. The experimental module called A, measuring $7.5 \times 2.6 \times 3.0$ m^3, has an opening (1.2×2.1 m^2) on each of the two long shear walls, see Fig. 1(*a*). Its geometry, material properties, and experimental results are previously reported [8], see also Fig. 1(*d*). Note that the dimensions of this module differ from those commonly used for residential construction, as illustrated in Fig. 1(*e*). Two larger typical modules (B and C) are introduced, see Fig. 1(*b*) and (*c*) with same material, module connection properties and double layer sheathing panels in the walls that are used in module A. Module B is an asymmetric module with a large opening (3.6×2.1 m^2) in the front shear wall while module C has the same opening as module B, supplemented with two door openings as shown in Fig. 1(*c*). Moreover, elastomer bricks of size $95 \times 95 \times 25$ mm^3 and type Sylomer® SR1200 are placed atop modules as inter-modular connections, see Fig. 1(*h*).

2.2. Design loads acting on a typical six-story

The design loads acting on typical multi-story buildings are calculated according to the principles presented in Eurocode 1 [51]. The six-story studied is taken from a central part of the building and no interaction is assumed between it and the side modules. Thus, the

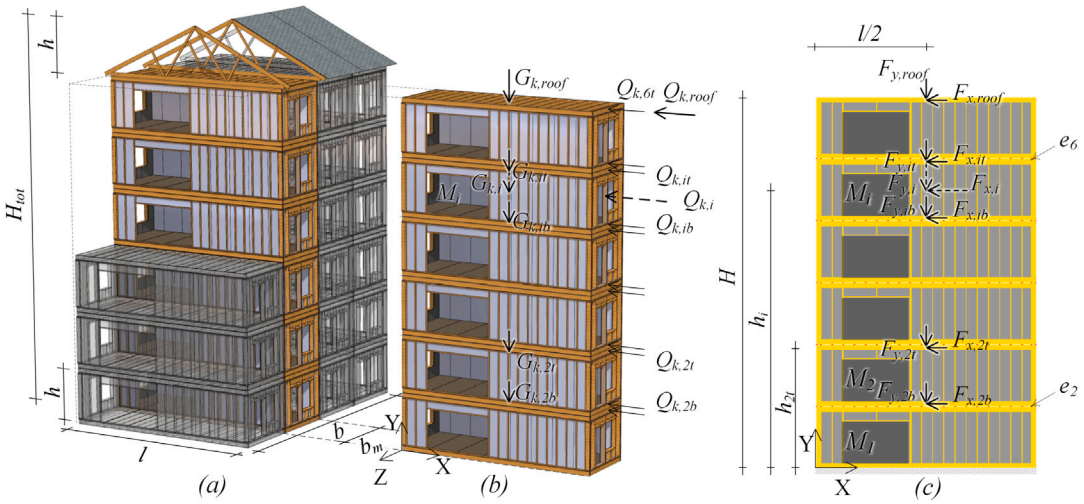


Fig. 2. (*a*) A typic.al 6 story modular building designed with module B, (*b*) gravity load $G_{k,i}$ and wind load $Q_{k,i}$ divided in to top ($_t$) and bottom ($_b$) diaphragm of the modules, and load from roof $G_{k,roof}$ and $Q_{k,roof}$ acting on the 6-story tower ($1 \le i \le 6$) taken from the central part of the modular building and (*c*) respective forces and resultant load height h_i in FE model.

wind load just acts normal to the short wall of each module, i.e., the long wall of a multi-story building is consisting of several such towers placed beside each other, see Fig. 2. The total height of the tower of six modules stacked on top of each other is $H_{tot} = 21$ m when the roof height is assumed to be $h = 3$ m at its highest point. For module B with width $b_m = 3.6$ m and length $l = 10.8$ m, while calculating for a basic wind velocity 26 m/s and terrain category 2, the peak velocity pressure $q_p(z) = 1.02$ kPa up to height $z = 10.8$ m and 1.2 kPa above that height. The characteristic wind load in each story (i) is calculated as:

$$Q_{k,i} = q_p(z) \cdot c_{pe} \cdot b_m \cdot h \tag{1}$$

where c_{pe} is the resultant external pressure coefficient from windward and leeward side for the wall, $c_{pe} = c_{pe,D} - c_{pe,E} = 1.3$.

The characteristic gravity load for each module $G_{k,i} \simeq 80$ kN and the roof weight $G_{k,roof} \simeq 20$ kN. The characteristic snow load is assumed to be $q_{k,sn} = 2.5$ kN/m^2 and the imposed load is considered to be $q_{k,im} = 2.0$ kN/m^2.

To calculate the unfavorable condition for slip and rigid body rotation (rocking) of the module, vertical loads are taken as minimum, hence the effect of snow and imposed load as the accompanying variable action is neglected. The characteristic combination where wind load is the leading action for the serviceability limit states Eq. 6.14a in [52] becomes

$$E_{d,i} = G_{k,i} + Q_{k,i} \tag{2}$$

The calculated design value of the effect of actions $E_{d,i}$ will be applied to each story of the simulation model, thus calculating arbitrary forces and stresses in individual fasteners or structural elements within the entire model. The characteristic loads $G_{k,i}$ and $Q_{k,i}$ are applied to each floor $1 \le i \le ns$, where ns = number of stories. Furthermore, the loads are divided into the top (t) and bottom (b) diaphragms of each module and therefore are denoted as $G_{k,it}$ and $G_{k,ib}$ for the gravity loads, and $Q_{k,it}$ and $Q_{k,ib}$ for the wind loads respectively for each story i. The weight of the roof $G_{k,roof}$ and the wind load acting on the roof $Q_{k,roof}$, see Fig. 2 are applied on the top diaphragm of the top floor.

2.3. Horizontal displacement in modules

The horizontal displacement at the top of a single module originates primarily from three deformation modes under the assumption of negligible bending of the module around the Z-axis as no vertical anchorage or tie-down is modeled e.g. [53], and they are:

- slip displacement in the inter-modular connection, u_{sl},
- rigid body rotation of the module, u_{br} and
- shear deformation of the module, u_{sh}.

The three deformation modes in a 2D model are illustrated in Fig. 3(*a*)–(*c*). The slip displacement u_{sl} can be further divided into shear deformation within the inter-modular connection and sliding (i.e., sliding between surfaces), but in this study these two components are summed together and referred to as slip. Previous research e.g. [4,16,20,54–56] has established that shear wall properties, particularly the sheathing-to-framing connection, are decisive for shear deformation. In addition, both displacement due to rigid body rotation and slip displacements are also important for multi-story analysis and they are mainly influenced by the inter-modular connection properties. The so-called story displacement or story drift, u_{tot} for a single module and one inter-modular connection (either the one on top or below the module), includes the combined effects of slip displacement u_{sl}, horizontal displacement due to rigid body rotation u_{br} and shear deformation u_{sh}, see displacement of point p_2^0 to p_2^a, p_2^b and p_2^c respectively in Fig. 3(*d*).

The story displacement u_{tot} is obtained from the FE results and divided first into the slip displacement u_{sl}, calculated as the difference between the displacements at the nodes above (p_1^0) and below (p_0) the elastomer brick model, see Fig. 3(*a*). To determine the shear deformation u_{sh}, the change in the diagonal length of the wall is used to form a trigonometric relation, see Eq. (3) and Fig. 3(*e*), assuming that the height h_0 and length l_0 of the module remain constant.

$$u_{sh} = \left(\sqrt{{d_c}^2 - {h_0}^2}\right) - l_0 \tag{3}$$

where d_c is the new diagonal length of the shear wall of the module, changed from the initial length d_0 due to the shear deformation of the module. Finally, the displacement due to rigid body rotation u_{br}

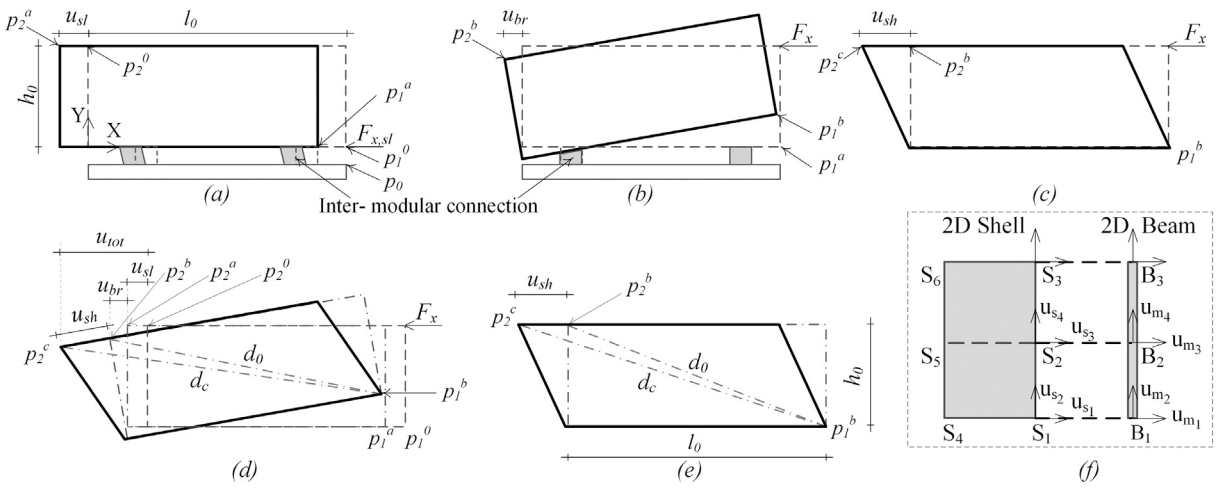


Fig. 3. Schematic illustration of types of displacement of the module, (*a*) slip displacement u_{sl}, (*b*) displacement due to rigid body rotation u_{br} including compression of elastomer bricks, (*c*) shear displacement u_{sh}, (*d*) story displacement u_{tot} combining those three types, (*e*) illustration for u_{sh} extracted from u_{tot} using trigonometry and (*f*) an example for kinematic constraints between a 2D shell element and a beam element.

is calculated by subtracting u_{sl} and u_{sh} from u_{tot}. In the results section, the story displacement is discussed along with the contribution from its three components. While determining the shear deformation u_{sh} from diagonal lengths, caution should taken for modules with large openings in their shear walls, as bending about the Z-axis might occur in those modules. However in the current study, the openings in the three modules studied are considered small, and thus bending deformations are neglected.

2.4. Adaptive and effective modeling

Given the significant number of unknowns in the entire model, model reduction techniques are used to maintain computational efficiency without compromising the reliability of the results. This section presents three methodologies implemented in an adaptive model to achieve this while maintaining the accuracy of the generated results within acceptable limits.

2.4.1. Kinematic constraints

The use of kinematic constraints in the FE method is a very efficient way to simulate certain types of (mechanical) constraints. It is used in the current work to impose constraints between different geometrical parts of the model, for instance in the inter-modular connections where DoFs on surface of certain 3D solid elements for the elastomer bricks are constrained to specific retained DoFs of some structural beam elements within the module. To achieve a high level of flexibility while keeping the computational cost low, the general method for imposing constraints on a system is to eliminate secondary DoFs, denoted as u_s, from the system DoFs u. For multi-point constraints, every linear relationship between different DoFs are defined as:

$$\sum_{i=1}^{n_{cDoF}} A_i \cdot u_i = g \tag{4}$$

where A_i are constant coefficients acting on the involved DoFs u_i, g is the so-called "gap value" for the constraint and n_{cDoF} is the number of DoFs involved in a specific constraint. The constraint equation is said to be homogeneous if the prescribed gap value $g = 0$, and non-homogeneous if $g \neq 0$ i.e., non zero [57]. For example $A_{m_1}u_{m_1} + A_{s_1}u_{s_1} = g_1$ is one linear constraint for $n_{cDoF} = 2$ between two nodal points B_1 of a 2D structural beam element as main (m) DoF and S_1 of a shell element as secondary (s) DoF, see Fig. 3(f). To kinematically constrain multiple nodal points involving several numbers of constraints n_{con} included in the system, the equations can be expanded into matrix form as:

$$\boldsymbol{A}_{con}\boldsymbol{u}_{con} = \boldsymbol{g}_{con} \tag{5}$$

$$\begin{bmatrix} A_{m_1} & 0 & 0 & 0 & \vdots & A_{s_1} & 0 & 0 & 0 \\ 0 & A_{m_2} & 0 & 0 & \vdots & 0 & A_{s_2} & 0 & 0 \\ 0 & 0 & A_{m_3} & 0 & \vdots & 0 & 0 & A_{s_3} & 0 \\ 0 & 0 & 0 & A_{m_4} & \vdots & 0 & 0 & 0 & A_{s_4} \\ \cdots & & & & \vdots & \cdots & & & \end{bmatrix} \begin{bmatrix} u_{m_1} \\ u_{m_2} \\ u_{m_3} \\ u_{m_4} \\ \cdots \\ u_{s_1} \\ u_{s_2} \\ u_{s_3} \\ u_{s_4} \\ \cdots \end{bmatrix} = \begin{bmatrix} g_1 \\ g_2 \\ g_3 \\ g_4 \\ \cdots \end{bmatrix} \tag{6}$$

where the matrix $\boldsymbol{A}_{con}$ contains all the coefficients operating on all the main and secondary DoFs in the constraints, $\boldsymbol{u}_{con}$ is a vector containing all DoFs (m) and (s) involved in the constraints and $\boldsymbol{g}_{con}$ is the gap vector. The matrix $\boldsymbol{A}_{con}$ can be partitioned into submatrices corresponding to the main ($\boldsymbol{A}^m_{con}$) and secondary ($\boldsymbol{A}^s_{con}$) DoFs as:

$$\boldsymbol{A}^m_{con}\boldsymbol{u}^m_{con} + \boldsymbol{A}^s_{con}\boldsymbol{u}^s_{con} = \boldsymbol{g}_{con} \tag{7}$$

Since $\boldsymbol{A}^s_{con}$ is a square and non-singular matrix, the secondary DoFs can be calculated as:

$$\boldsymbol{u}^s_{con} = -\boldsymbol{A}^{s-1}_{con}\boldsymbol{A}^m_{con}\boldsymbol{u}^m_{con} + \boldsymbol{A}^{s-1}_{con}\boldsymbol{g}_{con} = \boldsymbol{A}^{tr}_{con}\boldsymbol{u}^m_{con} + \boldsymbol{g}^{tr}_{con} \tag{8}$$

where $\boldsymbol{A}^{tr}_{con} = -\boldsymbol{A}^{s-1}_{con}\boldsymbol{A}^m_{con}$ is the transformation matrix and $\boldsymbol{g}^{tr}_{con} = \boldsymbol{A}^{s-1}_{con}\boldsymbol{g}_{con}$ is the reduced gap vector. To apply these relationships to the global equation system with total number of DoFs n_{dof}, the transformation matrix need to be expanded (exp) to the full set of DoFs as:

$$\boldsymbol{A}^{tr}_{exp} = \begin{bmatrix} 1 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & \boldsymbol{A}^{tr}_{con(1,1)} & \cdots & 1 & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & \boldsymbol{A}^{tr}_{con(2,2)} & \cdots & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 & \ldots & 1 \end{bmatrix}_{n_{dof}\times(n_{dof}-n_{con})} \tag{9}$$

Using the global stiffness matrix $\boldsymbol{K}$ of size $n_{dof} \times n_{dof}$, together with the global load vector $\boldsymbol{f}$ and the gap vector $\boldsymbol{g}$ of sizes $n_{dof} \times 1$, the reduced global stiffness matrix $\boldsymbol{K}_{red}$ of size $(n_{dof} - n_{con}) \times (n_{dof} - n_{con})$ and the

Table 1
Material data used for beam, shell and solid elements in the model.

Component	Material	Type	Material parameters [MPa]	ν
Beam	Timber (C24)	Orthotropic	$E_1 = 11\,000$, $G_{12} = G_{13} = 690$	0.3
Shell	OSB (type 3)	Orthotropic	$E_1 = 3800$, $E_2 = 3000$, $G_{12} = 1080$	0.3
Shell	Gypsum (Gyproc F2)	Isotropic	$E = 2500$	0.3
Shell	CLT (3 layers)	Orthotropic	$E_1 = 11\,000$, $G_{12} = 690$	0.3
Solid	Elastomer brick	Isotropic	$E_e = 9.3$	0.3

reduced load vector $\boldsymbol{f}_{red}$ of size $(n_{dof} - n_{con}) \times 1$ becomes

$$\boldsymbol{K}_{red} = \boldsymbol{A}_{exp}^{trT} \boldsymbol{K} \boldsymbol{A}_{exp}^{tr} \tag{10}$$

$$\boldsymbol{f}_{red} = \boldsymbol{A}_{exp}^{trT} (\boldsymbol{f} - \boldsymbol{K}\boldsymbol{g}) \tag{11}$$

The reduced equation system is then formulated as

$$\boldsymbol{K}_{red} \boldsymbol{u}_m = \boldsymbol{f}_{red} \tag{12}$$

and the global main DoFs can be solved following the conventional FE methodology. After solving the main displacements the total displacement vector $\boldsymbol{u}$, including the main and secondary DoFs can be obtained as:

$$\boldsymbol{u} = \boldsymbol{A}_{exp}^{tr} \boldsymbol{u}_m + \boldsymbol{g} \tag{13}$$

For further information on modeling of kinematic constraints, the reader is referred to, e.g. [57]. The drafted constraint modeling is computationally efficient and flexible. However, there are also other methods available, such as the penalty method and the use of Lagrangian multipliers.

2.4.2. Substructuring

An efficient and well established methodology to reduce the number of DoFs in an equation system is to use Guyan reduction [48] often referred to as substructuring. In matrix form, the standard FE-formulation for a static boundary problem can be expressed as

$$\boldsymbol{K}\boldsymbol{u} = \boldsymbol{f} \tag{14}$$

where $\boldsymbol{K}$ is the stiffness matrix, $\boldsymbol{u}$ is the displacement vector and $\boldsymbol{f}$ is the force vector. In such an equation system, the stiffness matrix could be arranged so that it allows elimination of a selected set of DoFs. These eliminated DoFs will be secondary (s) and the remaining DoFs are referred to as retained DoFs (r). The latter are available for loading and boundary conditions, attaching elements and extracting output directly. The rearranged FE-formulation would then take the following form:

$$\begin{bmatrix} \boldsymbol{K}_{rr} & \boldsymbol{K}_{rs} \\ \boldsymbol{K}_{sr} & \boldsymbol{K}_{ss} \end{bmatrix} \begin{bmatrix} \boldsymbol{u}_r \\ \boldsymbol{u}_s \end{bmatrix} = \begin{bmatrix} \boldsymbol{f}_r \\ \boldsymbol{f}_s \end{bmatrix} \tag{15}$$

where $\boldsymbol{u}_r$ represents the retained DoFs and $\boldsymbol{u}_s$ the secondary DoFs. The secondary DoFs can be obtained from the second equation of the matrix system as

$$\boldsymbol{u}_s = \boldsymbol{K}_{ss}^{-1} \boldsymbol{f}_s - \boldsymbol{K}_{ss}^{-1} \boldsymbol{K}_{sr} \boldsymbol{u}_r \tag{16}$$

If this equation is used to substitute the secondary DoFs in Eq. (15) the following system of equations is obtained

$$\left[\boldsymbol{K}_{rr} - \boldsymbol{K}_{rs} \boldsymbol{K}_{ss}^{-1} \boldsymbol{K}_{sr}\right] \boldsymbol{u}_r = \boldsymbol{f}_r - \boldsymbol{K}_{rs} \boldsymbol{K}_{ss}^{-1} \boldsymbol{f}_s \tag{17}$$

which in short can be expressed as:

$$\boldsymbol{K}_{red} \boldsymbol{u}_r = \boldsymbol{f}_{red} \tag{18}$$

By using the Guyan reduction method outlined above, the secondary DoFs are condensed, and the remaining geometrical entity is referred to as a superelement. In the FE package Abaqus, this element can constitute, e.g. an arbitrary module, and that is how the reduction technique is used in the next part of the paper. Once the retained DoFs are solved with equation system in Eq. (17), the secondary ones may also be achieved by using Eq. (16).

2.4.3. Modeling of the modular tower

When stacking modules to simulate an entire modular tower, the inter-modular connections become one of the key elements in the building. To reduce the computational cost without reducing the model accuracy for elastic deformations, the current modeling approach uses the previously described substructuring technique. In the following simulations, the bottom module M_1 is modeled with standard non-linear FE-formulation to simulate the possible non-linear shear behavior within the module, while the remaining modules (M_2–M_6) are simulated as five separate substructures. It should be noted that the model is flexible and fully parameterized in the sense that any number of modules can be modeled with the standard FE method; however, it has been deemed sufficient to simulate only one here in that manner.

Fig. 4(a) shows a model geometry of a six-story modular tower composed of similar modules (module A) as used in a full-scale experimental module tests described in [8] and illustrated in Fig. 1(d). The modules are modeled with 3D beam elements for timber frame members and shell elements for sheathing panels. Material properties of the elements used are tabulated in Table 1. Moreover, mechanical sheathing-to-framing connections between the sheathing panels (single or double) and the frame elements used in shear walls, transverse walls and floor diaphragms are modeled as coupled elasto-plastic springs and the characteristics are illustrated in Fig. 5(a) (see also module A [9]). The slip modulus is $k_0 = 530$ kN/m, initial yield force P_0 is 200 N and second yield force P_1 is 400 kN. Intra-modular connections are the mechanical connections between the shear walls and transverse walls, and between the walls and floor diaphragms. They are modeled using uncoupled non-linear elastic springs with intra-modular slip modulus, $k_i = 20$ MN/m and initial yield force $P_i = 3.2$ kN, see Fig. 5(b).

The substructure modules have the same geometry and design as the bottom module, but their behavior is assumed to be purely linear. The retained DoFs in the substructure modules are located at the top and bottom of the edge beams of the floor elements, corresponding to locations of the elastomer bricks as shown for the bottom module in Fig. 4(b) and for other modules in Fig. 4(c) [8]. All other DoFs apart from the retained DoFs in the substructure module are reduced and not included in the simulations, resulting in a significantly small system matrix. Fig. 4(d) illustrates in detail how the inter-modular connection between an edge beam in the standard module M_1 and the retained nodal points in the substructure module M_2 is modeled.

Since these modules are of platform type design, their floor elements are placed above and below the walls. The bottom surfaces of the elastomer bricks are fully constrained to the nodal points (Node 1) at the edge beams in the top floor elements beneath them using kinematic constraints. The elastomer bricks themselves are modeled with 3D solid elements and their top surfaces are coupled to reference points, also through kinematic constraints. Non-linear coupled springs k_{ex}, k_{ey} and k_{ez} are used to connect these reference points and the retained nodal points (Retained node 2) at the edge beams in the bottom floor elements above. The material behavior of the coupled springs and the solid elements for the elastomer bricks have been defined in Fig. 5(b) and (c) (see also [9]) and they are used to simulate shear deformations and non-linear friction based slip deformations in the inter-modular connections. The vertical compressive stiffness of the coupled springs is $k_{ey,c} = 5$ MN/m while the tensile stiffness is $k_{ey,t} = 1$ N/m $\simeq 0$, see Fig. 5(b). This suggests that the elastomer bricks are unaffected when tensile loads act on it. In other words, if the vertical compression forces

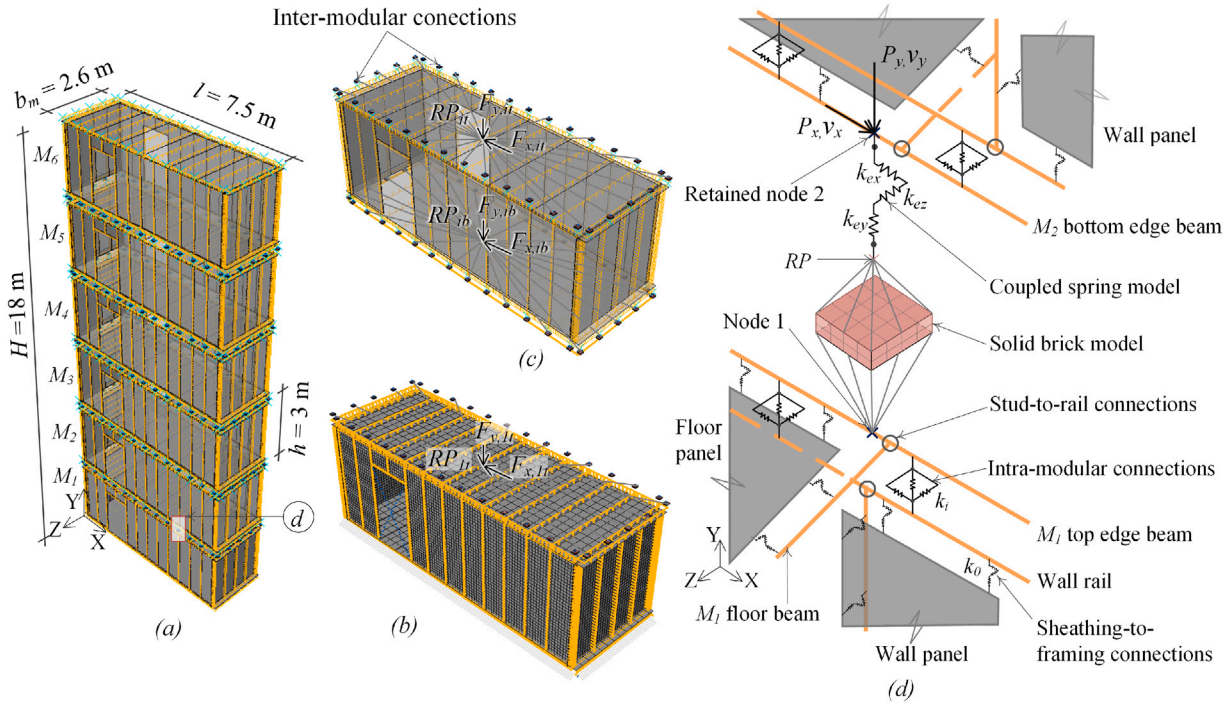


Fig. 4. (*a*) Illustration of a modeling approach for a six-story modular tower (module A) using standard FE-formulation (Eq. (14)) for the bottom module (M_1) and substructure formulation (Eq. (18)) for modules (M_2–M_6), location of inter-modular connections and reference points (RP) used to load the elastomer bricks for (*b*) the bottom module M_1 and (*c*) a typical substructure module M_i, and (*d*) detailed illustration of the inter-modular connections (solid and spring model) between M_1 and M_2.

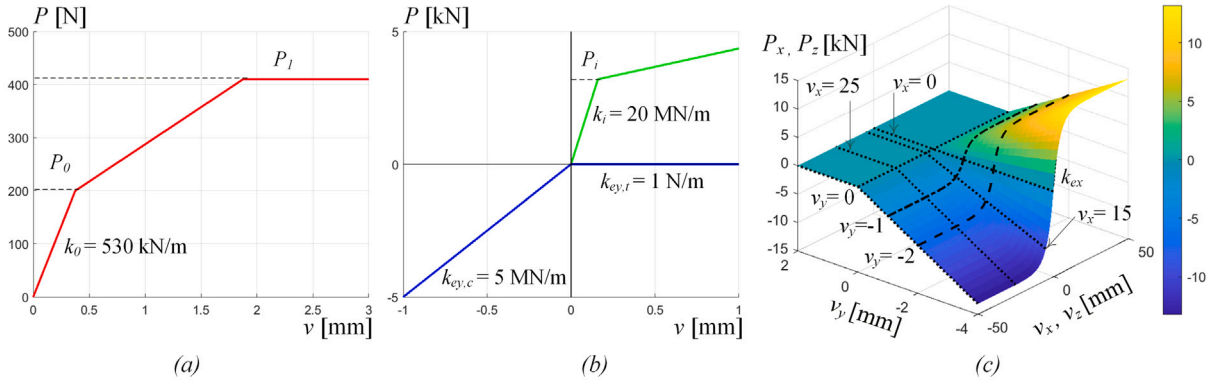


Fig. 5. Load–displacement behavior for (*a*) coupled sheathing-to-framing connections, (*b*) uncoupled intra-modular connections (slip modulus k_i) and inter-modular connections (vertical slip modulus k_{ey}) and (*c*) coupled elasto-plastic spring connection relation between $P_x - v_x$ or $P_z - v_z$ and v_y.

are reduced due to local uplift, the elastomer bricks return elastically to their initial position and allow uplift of the modules above it and as a consequence, no shear force transmission occurs. Furthermore, the shear force transfer behavior depends on the magnitude of the compressive displacement of the elastomer bricks, see Fig. 5(*c*). The compression side of the horizontal force along X-axis P_x and vertical displacement v_y (negative values of v_y) is modeled linear but the relation between P_x and horizontal displacement v_x is asymptotic, based on the experimental results.

An experimental verification of the standardized modeling work of individual modules including inter-modular connections has previously been performed [9,47] but validation of the substructure models will be presented in the following section. The simulation results will be compared to the experimental results described in [8].

To represent a typical loading scenario for multi-story building, design load is applied, based on the characteristic load combination outlined in Section 2.2 for a typical six-story building. The loading sequence in the simulations begins with applying the gravity load F_y, followed by the wind loads F_x applied to each story (i) of the tower. For the substructure modules, all loading is applied on both the top (t) and the bottom (b) diaphragm of each module through reference points (RP_{it} and RP_{ib}) that are connected to all retained nodal points in the modules at the story levels ($2 \le i \le 6$), see Fig. 4*(c)*. For the bottom module, loads at the top diaphragm are applied through a reference point RP_{1t}, that is kinematically constrained to all nodal points on the edge beams of the top floor element located below the elastomer bricks, see Fig. 4*(b)*. The horizontal and vertical forces applied at the top and the bottom diaphragm of each module are denoted $F_{x,i}$ and $F_{y,i}$ respectively.

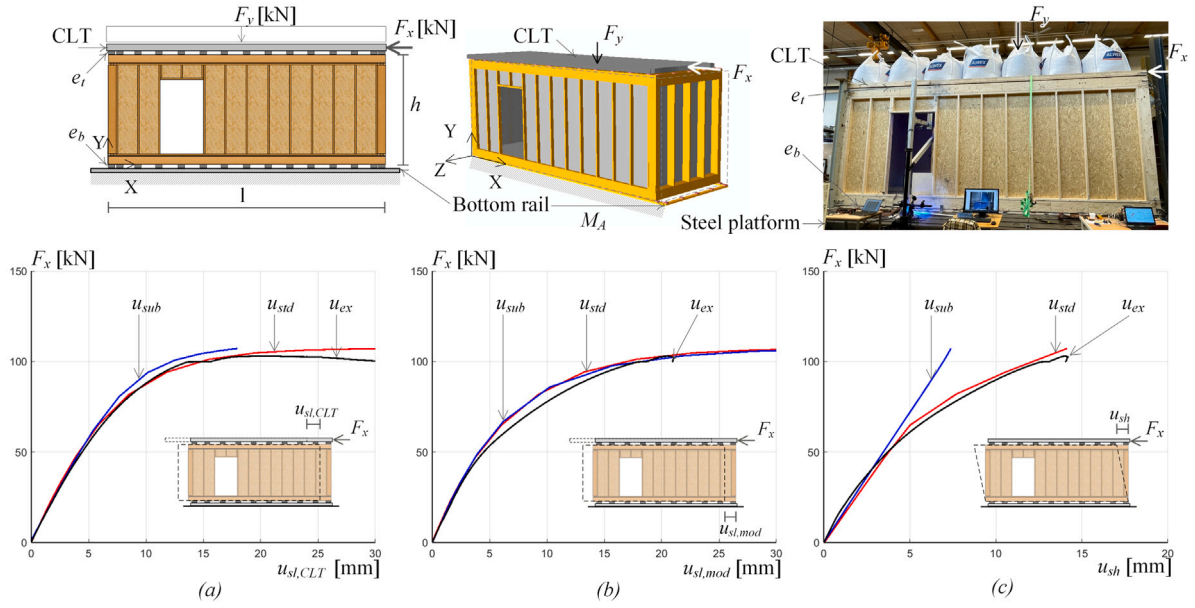


Fig. 6. Comparison between three force–displacement curves representing experimental results ($_{ex}$), modeling results with standard FE-formulation ($_{std}$) and substructure formulation ($_{sub}$) for relative slip deformation in the connection between (a) the CLT panel and the module top, $u_{sl,CLT}$ due to top elastomer bricks layer e_t (b) the bottom rails and the module bottom $u_{sl,mod}$ due to bottom elastomer bricks layer e_b and (c) relative shear displacement u_{sh}.

3. Simulations of structures consisting module A

This section presents numerical and experimental results of a single test model of module A and some simulation results of the modular tower where both substructure and standard FE-formulation are used.

3.1. Experimental verification of a single test module including inter-modular connections

To verify the new substructure formulation, some results are compared to previously obtained results from a standard FE model in [9] and with experimental test results in [8]. The simulated module is complemented with a CLT panel at the top and support rails at the bottom, similar to the full-scale experimental tests. The material properties used in the modeling are enlisted in Table 1. The connections between the CLT panel and module, and between the bottom rails and the module are identically modeled as inter-modular connections with elastomer bricks, regardless of whether the module is simulated with substructure or standard FE-formulation.

Fig. 6 demonstrates three sub-figures for comparison, one for the relative slip displacement within the friction based (due to elastomer bricks layer e_t) CLT-to-module connection $u_{sl,CLT}$, one for the relative slip displacement in the module-to-bottom rail connection $u_{sl,mod}$, and finally one for the relative displacement caused by shear displacement u_{sh} in the module itself. Each sub-figure shows three force–displacement curves representing the experimental results (ex) and results from the two simulation applications with substructures (sub) and with standard FE-formulation (std).

The comparison reveals good agreement between the three slip displacements for both the CLT and the module, with accuracies of approximately 95% and 99% respectively, see Fig. 6(a) and (b). The response is initially linear, followed by a non-linear part which ultimately ends in a plateau region indicating a pure sliding between the elastomer bricks and the wood material. The agreement between the two simulation methods was expected since both the models use the same modeling approach to simulate the inter-modular connections. However, since the substructure model is limited to purely linear-elastic behavior, it cannot capture the non-linear displacement response for the shear displacement u_{sh}, caused by the elasto-plastic deformations of the sheathing-to-framing connections in the module. Fig. 6(c) shows that the shear displacement response remains linear up to a load level of $F_x \simeq 50$ kN. This means that the substructure model can be used to obtain the same results as those in standard FE-formulation for module simulation up to that load level.

While providing approximately 95% accuracy in predicting the shear displacement u_{sh} up to the load level $F_x \simeq 50$ kN, there is significant reduction in computation time when the substructure model is used. CPU time taken by substructure generation is $t_{sub1} = 148$ s and by substructure analysis of the single module is $t_{sub2} = 1720$ s whereas, the CPU time taken by the same computer for the standard FE analysis of that module is $t_{std} = 5930$ s. This 68.5% reduction in computational time for analyzing the module is a trade-off between computational efficiency and accuracy.

3.2. Simulation of multi-story building

The simulation model presented is used to simulate a six-story modular tower composed of module A, see Fig. 7. Vertical F_y and horizontal F_x loads are applied with characteristic loads [52] described in Fig. 2 and Table 2. The dimension, loads, story drift and simulated geometry configuration of the building (using a magnification factor of 10) is illustrated in Fig. 7(a) and (b). The horizontal roof load by wind acting on the top module $F_{x,roof}$, wind load acting on the top $F_{x,it}$ and bottom $F_{x,ib}$ diaphragm of each story i, as well as the vertical roof load by wind and self-weight of roof acting on the top module $F_{y,roof}$ and the vertical load caused by self-weight of each module divided into the top $F_{y,it}$ and bottom $F_{y,ib}$ diaphragm of each story i are all tabulated in Table 2. The roof loads are added on the top diaphragm of the top module and the top and bottom loads applied in respective diaphragms of each story. There is a cumulative effect of the applied loads from top to bottom so that the bottom story is the most loaded both in vertical and horizontal directions.

A story drift $u_{tot,i}$ is a horizontal displacement of an individual module at story i including inter-modular connection between them. This displacement measure is of interest for serviceability limit state design, see Fig. 7(c). The total horizontal displacement u_{tot} is the sum of all story drift $u_{tot,i}$, which in turn is the combination of shear displacement $u_{sh,i}$, horizontal displacement caused by rigid body rotations

Table 2
Loads and resultant load heights on top and bottom diaphragm of each module used to simulate the tower.

Module (i)	Load position	Gravity load $F_{y,i}$ ($G_{k,i}$)	[kN]	Wind load $F_{x,i}$ ($Q_{k,i}$)	[kN]	Load height h_i [m]
M_1	Top of module M_1	$F_{y,1t}$	40	$F_{x,1t}$	7.5	3
M_2	Top and bottom	$F_{y,2t} + F_{y,2b}$	80	$F_{x,2t} + F_{x,2b}$	15	4.5
M_3		$F_{y,3t} + F_{y,3b}$	80	$F_{x,3t} + F_{x,3b}$	15	7.5
M_4		$F_{y,4t} + F_{y,4b}$	80	$F_{x,4t} + F_{x,4b}$	15	10.5
M_5		$F_{y,5t} + F_{y,5b}$	80	$F_{x,5t} + F_{x,5b}$	15	13.5
M_6		$F_{y,6t} + F_{y,6b}$	80	$F_{x,6t} + F_{x,6b}$	15	16.5
Roof	Top of module M_6	$F_{y,roof}$	20	$F_{x,roof}$	15	18
Total			460		97.5	

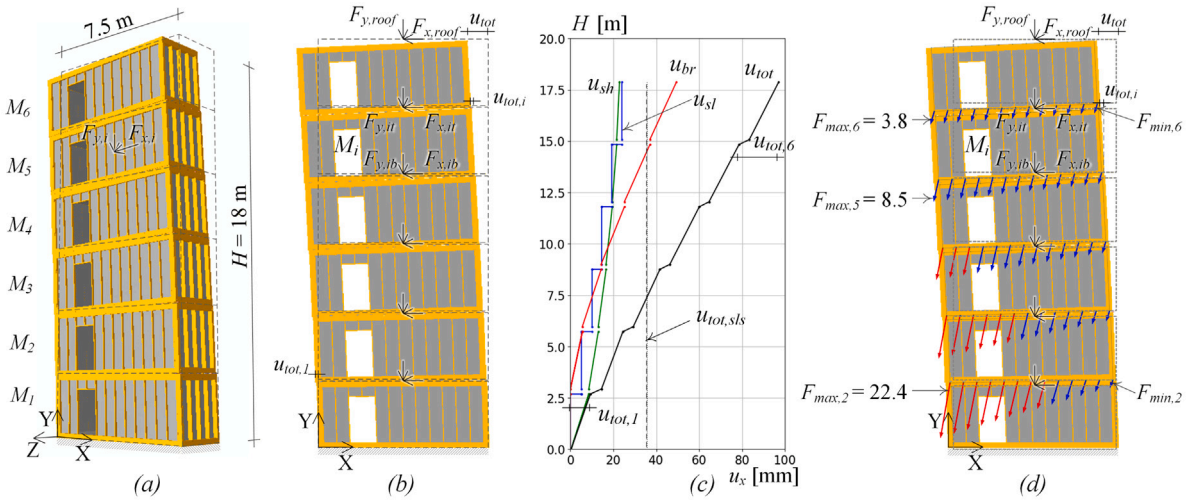


Fig. 7. Simulated modular tower composed of module A (a) dimensions, (b) loads and deformed shape of the modules, (c) components of story drift $u_{tot,i}$ versus height of tower and allowable horizontal drift $u_{tot,sls}$ and (d) internal forces (in kN) in elastomer bricks within the tower. For interpretation of the references to color in this figure, the reader is referred to the web version of this article.

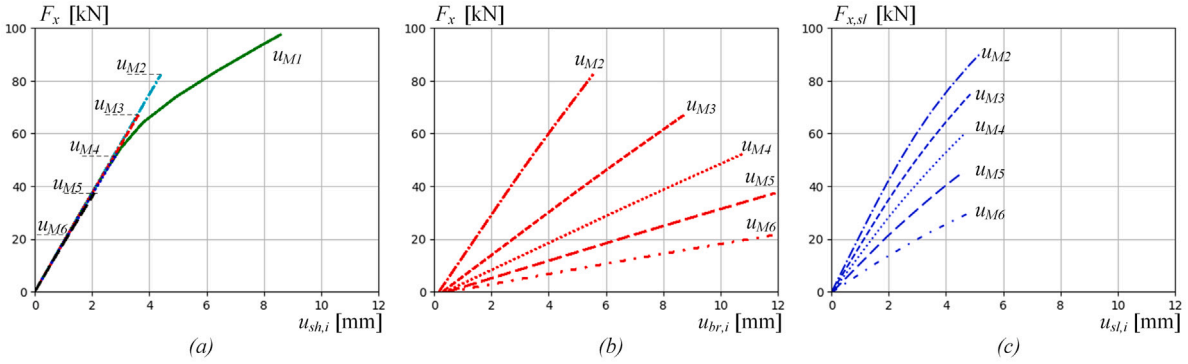


Fig. 8. A force displacement diagram for the sub-displacements in the story drift $u_{tot,i}$ (a) shear displacement of an individual module $u_{sh,i}$, (b) displacement caused by rigid body rotation $u_{br,i}$ and (c) inter-modular slip displacement $u_{sl,i}$.

$u_{br,i}$ and slip in inter-modular connections $u_{sl,i}$ for each module. For the serviceability limit state design, the limit value for the story drift is typically set to $u_{sls} \leq h/500$ [58–61] . Applying this for a single module of height $h = 3$ m, $u_{sls,i} = 6$ mm while for a six-story tower of height $H = 18$ m, $u_{tot,sls} = 36$ mm.

Fig. 7(d) demonstrates the internal forces F acting on the elastomer bricks as a result from both vertical and horizontal loads acting on the modules. In all the modules, these internal forces increase linearly from right to left because of the wind load. The forces F are illustrated by a blue vector for $F \leq 10$ kN and a red vector for $F > 10$ kN. The static range of use of the elastomer bricks defined in this model is 1.2 MPa which corresponds to the static load $F_e \simeq 10.8$ kN for the brick geometry used here. The vector plot shows that the elastomer bricks of only two upper modules M_5 ($F_{max,5} = 8.5$ kN) and M_6 ($F_{max,6} = 3.8$ kN) are within the static range of the elastomer brick type. However, elastomer bricks of lower modules should be adjusted to the correct static range to ensure proper behavior. The maximal internal force $F_{max,2} = 22.4$ kN is found at the left side of the module M_2, and the load exceeds the static range by more than a factor of two.

To study the story drift $u_{tot,i}$ further, Fig. 8 visualizes it as three force–displacement diagrams showing its three different displacement components explained in Fig. 3(d). The shear displacement $u_{sh,1}$ (for module M_1) is linear up to $F_x \simeq 50$ kN and thereafter non-linear up to a force value of $F_x = 97.5$ kN, see Fig. 8(a). Since the bottom module M_1 is exposed to the highest loads and behaving strongly non-linear, it is necessary to use the standard FE-formulation to simulate that module. Furthermore, modules M_2 and M_3 are also loaded above the plasticity onset while modules M_4–M_6 are loaded only at the linear elastic

Table 3
Contribution from displacement components on story drift $u_{tot,i}$ for modules in the tower composed of module A.

Module	F_y [kN]	F_x [kN]	$u_{sh,i}$ [mm]	$u_{br,i}$ [mm]	$F_{x,sl}$ [kN]	$u_{sl,i}$ [mm]	$u_{tot,i}$ [mm]	$u_{sh,i}$ [%]	$u_{br,i}$ [%]	$u_{sl,i}$ [%]
M_1	460	97.5	8.6	–	–	–	8.6	100	–	–
M_2	420	82.5	4.4	5.6	90	5.2	15.1	29.1	36.7	34.1
M_3	340	67.5	3.6	8.8	75	4.9	17.3	21.1	50.8	28.1
M_4	260	52.5	2.9	10.8	60	4.7	18.3	15.7	59.0	25.4
M_5	180	37.5	2.1	11.9	45	4.5	18.5	11.2	64.3	24.5
M_6	100	22.5	1.3	12.3	30	4.8	18.4	6.9	66.9	26.2
Cumulative			22.8	49.4		24.0	96.2			

part. In this study, the modules M_2 to M_6 are modeled with linear substructure modeling. This simplification is deemed acceptable here, since it contributes to increased computational efficiency necessary to get reasonable computational time for the modular tower simulation.

On the other hand, the horizontal displacement due to rigid body rotation of each modules $u_{br,i}$ increases with increasing floor height, see Fig. 8(*b*). Note that module M_1 is assumed to be fixed to the ground, i.e., $u_{br,1} = 0$ whereas the other modules have significant rigid body rotations which results in slightly non-linear increase with floor height and getting $u_{br,6} = 12.3$ mm, while the horizontal load F_x is only 22.5 kN.

Furthermore, the slip displacement $u_{sl,i}$ is quite similar for all the inter-modular connections $u_{sl,i} \simeq 5$ mm, see Fig. 8(*c*). This is due to the linear increment of horizontal F_x and vertical F_y loads from top to bottom modules and the fact that the horizontal force P_x versus vertical displacement v_y relation is defined linear in compression for the inter-modular connections (elastomer bricks), see [9] and Fig. 5(*c*). Note that the force $F_{x,sl}$ causing slip displacement is larger than the force F_x causing shear displacement (by 7.5 kN) since it includes the load $F_{x,ib}$ acting on the lower half of the module, see Fig. 3(*a*) and Table 2.

To further scrutinize how these three displacement components influence the story drift $u_{tot,i}$, Table 3 lists force, displacement and percentage values for all the modules. Observe that the shear displacements $u_{sh,i}$ decrease from module M_2 to top module M_6, the share decreasing from 29.1% to 6.9% whereas the displacement due to rigid body rotation u_{br} increases its share from 36.7% to 66.9%. The slip displacement u_{sl} ranges from 5.2–4.5 mm with decreasing share from module M_2 to M_5. The total horizontal displacement at the top floor $u_{tot} = 96.2$ mm, 2.67 times the recommended limit value $u_{tot,sls}$.

Furthermore, to check the stability of the tower against overturning, the factor of safety is calculated. The overturning moment calculated at the origin in Fig. 7 becomes $M_{o,A} = \Sigma(F_{x,i} \cdot h_i) = 1080$ kN m whereas the resisting moment is $M_{r,A} = \Sigma(F_{y,i} \cdot l/2) = 1725$ kN m, see Table 2 for $F_{x,i}$, $F_{y,i}$ and h_i. Thus the factor of safety against overturning for tower composed of module A is $\eta_A = M_{r,A}/M_{o,A} = 1.59$, which is acceptable for the design wind load.

4. Results and discussions

This section presents and discusses results from simulations of a six-story tower composed of module B for the global behavior of the tower and local structural behavior of the bottom module.

4.1. Multi-story tower composed of module B

Similarly to the previous section, the bottom module of the tower is simulated with the standard FE-formulation and the five modules stacked above are modeled with substructure formulation, see Fig. 9(*a*). The bottom module M_1 is fixed to the ground, and an assembly of similar substructure modules M_2–M_6 is stacked with inter-modular connections representing elastomer bricks in between them. The loads given in Table 2 are applied, first in vertical ($F_{y,i}$) and then in horizontal ($F_{x,i}$) directions, through the reference points RP_i at the top (t) and bottom (b) diaphragm of each modules. The RP_i are kinematically constrained in all DoFs to the retained nodal points of each diaphragm for the substructure modules and to the nodal points on the edge beams of the top floor element for the bottom module as described in Section 2.4.3. Fig. 9*(a)* shows the deformed state of the six-story and Fig. 9*(b)* and *(c)* show the deformed shear walls (front face) under vertical loading and combined loading respectively, with a magnification factor 10.

To determine the racking strength of the tower, the horizontal wind loads $F_{x,roof}$, $F_{x,it}$ and $F_{x,ib}$ are linearly increased while the vertical loads $F_{y,i}$ are kept constant. The force–displacement curves in Fig. 9(d) show the relation between the applied total horizontal force F_x, story drift $u_{tot,i}$ and total vertical force F_y acting on each module of the tower. It illustrates that the cumulative forces $F_{x,1}$ and $F_{y,1}$ for module M_1 is 225 kN and 460 kN respectively, whereas $F_{x,6} = 52$ kN and $F_{y,6} = 100$ kN for module M_6. The plot reveals that there is a significant increase in story drift $u_{tot,1}$ of the bottom module M_1 when the horizontal force reaches $F_{x,1} = 200$ kN. This is caused by an increase in non-linear shear displacement in module M_1. The story drift $u_{tot,1}$ increases asymptotically for loads $F_{x,1} \geq 225$ kN indicating this as the maximum force $F_{x,max}$ or the racking strength of the module M_1. The racking stiffness of the module M_1 is calculated as $K_1 = (F_{x,max,40}\text{-}F_{x,max,20})/(u_{sh,1,40}\text{-}u_{sh,1,20}) = 23$ kN/mm. The stiffness and racking strength of the module are the same for modules in all story due to their similar design. The fact that the racking strength is reached for module M_1 implies that the racking strength of the six-story tower is reached at the load level $F_{x,max} = 225$ kN.

Fig. 9*(b)* and *(c)* also show the force vectors between the modules which represent internal forces F acting in the elastomer bricks. The vectors in blue represent $F \leq 10$ kN and in red $F > 10$ kN as discussed in previous section. During pure vertical loading $F_{y,i}$ on the modules, the forces F are oriented vertically and relatively evenly distributed on each floor. Upon application of the horizontal load $F_{x,i}$, the forces F increase at the left part while it decrease at the right part. At the top of the tower, the difference between left and right parts is less due to the limited overturning moment from applied horizontal loads.

The force F for two elastomer bricks below the module M_2 and M_6 are reported in Fig. 9*(b)* under vertical loading when $F_{y,1} = 460$ kN, and in Fig. 9*(c)* under combined loading when the horizontal force $F_{x,1}$ reached $F_{x,max} = 225$ kN on the module M_1. The forces on the elastomer bricks F_2 is 9.5 kN at the left end and 8.1 kN at the right end of the module M_2 during vertical loading. After application of the horizontal load, the maximum force on the elastomer brick is $F_{max,2} = 27.1$ kN at the left end of the module M_2 while the minimum force at the right end $F_{min,2} = 0$ indicating contact free elastomer bricks due to uplift action of the module. This shows an increment in forces by a factor of 2.8 when the racking strength of the tower is reached due to shear failure in bottom module M_1.

The time history of the internal forces F for the selected elastomer bricks are illustrated in the sub-figures (f) - (i). The elastomer brick at the left end below the module M_2 has experienced an increment of $\Delta F_2 = 17.6$ kN. Furthermore, for module M_6, the force increases by $\Delta F_2 = 2$ kN at the left end and decreases by 0.8 kN at the right end. The horizontal displacement of modules M_2 and M_6 are illustrated with a magnification factor 10.

An interesting study for the tower is the story drift $u_{tot,i}$ to be compared with the limit value $u_{sls,i} = 6$ mm when the force F_x reaches

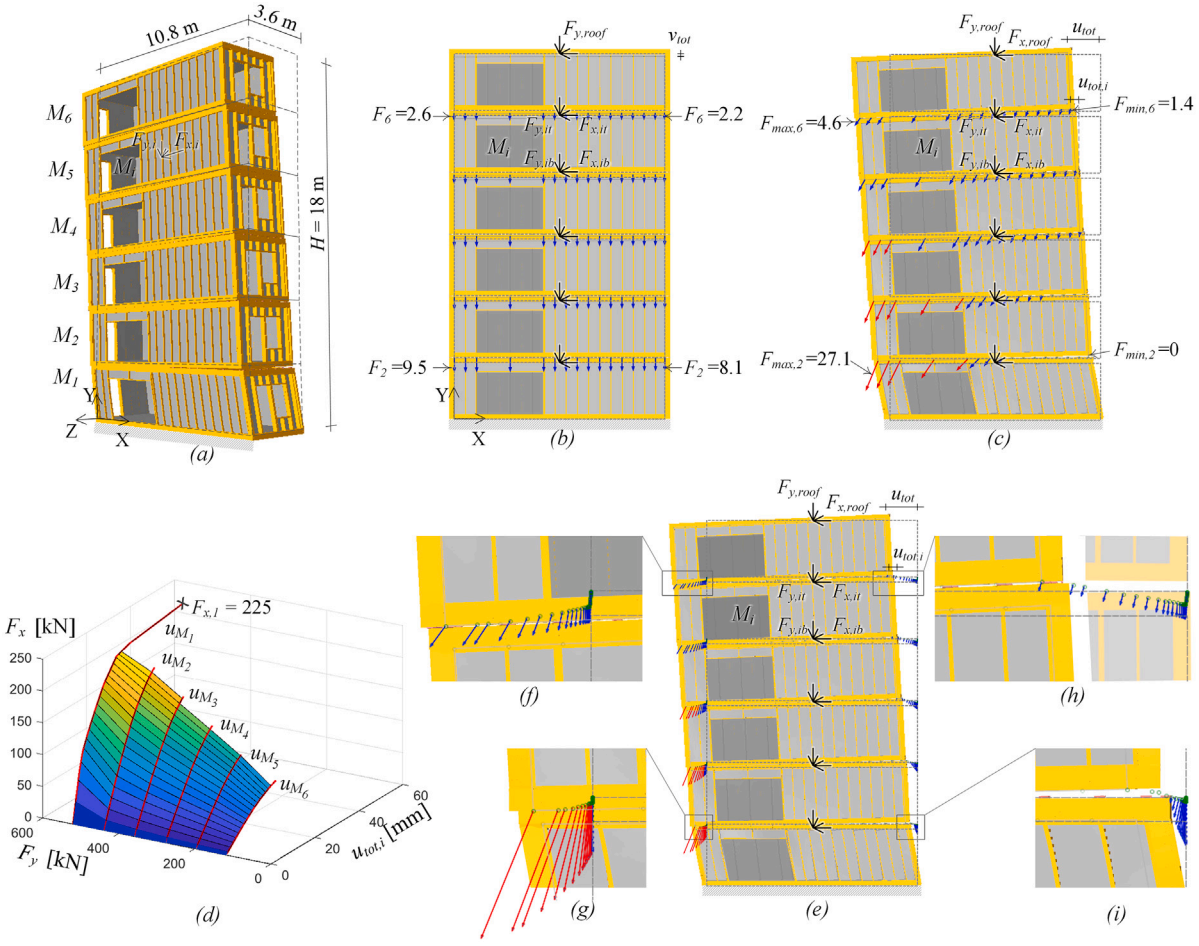


Fig. 9. (*a*) Deformed model of the tower composed of module B, forces (in kN) on selected elastomer bricks during (*b*) vertical loading and (*c*) combined loading, (*d*) force–displacement diagram showing horizontal force F_x, vertical force F_y and story drift $u_{tot,i}$, (*e*) force history of selected elastomer bricks during combined loading and (*f*) – (*i*) enlarged force history for some elastomer bricks when the maximum horizontal force $F_{x,max}$ = 225 kN in module M_1. For interpretation of the references to color in this figure, the reader is referred to the web version of this article.

Table 4
Story drift $u_{tot,i}$ when $F_x = Q_{k,1}$ = 97.5 kN at the bottom module in the tower composed of module B.

Module	F_y [kN]	F_x [kN]	$u_{sh,i}$ [mm]	$u_{br,i}$ [mm]	$u_{sl,i}$ [mm]	$u_{tot,i}$ [mm]	$u_{tot,i}/u_{sls,i}$ [–]
M_1	460	97.5	4.5	–	–	4.5	0.7
M_2	420	82.5	3.4	2.0	2.3	7.7	1.3
M_3	340	67.5	2.8	3.2	2.2	8.2	1.4
M_4	260	52.5	2.1	4.0	2.1	8.2	1.4
M_5	180	37.5	1.5	4.4	2.2	8.1	1.4
M_6	100	22.5	0.9	4.6	2.9	8.4	1.4
Cumulative			15.2	18.2	11.7	45.1	

the characteristic load $Q_{k,1}$ = 97.5 kN at the bottom module, see Table 4. The story drift $u_{tot,1}$ for the bottom module M_1, due to only the shear displacement $u_{sh,1}$, is within the limit $u_{sls,i}$, whereas module M_2 has $u_{sh,2}$ = 3.4 mm, displacement due to rigid body rotation $u_{br,2}$ = 2 mm and slip displacement $u_{sl,2}$ = 2.3 mm, resulting in the story drift $u_{tot,2}$ 1.3 times the limit value. For modules $i > 2$, $u_{br,i}$ has a major effect on $u_{tot,i}$ as $u_{sh,i}$ is decreasing along with story rise and $u_{sl,i}$ is almost the same, resulting in the story drift 1.4 times the limit value. In total, slip displacement has 26%, shear displacement has 33.7% and displacement due to rigid body rotation has the most 40.3% contribution in the total horizontal displacement u_{tot}.

Moreover, the overturning moment $M_{o,B}$ = 1080 kN m and the resisting moment is $M_{r,B}$ = 2484 kN m. Thus, the factor of safety against overturning for the tower composed of module B is η_B = 2.3, indicating satisfactory global stability under the design wind load.

4.2. Bottom module of the module B tower

For the standard FE model of bottom module M_1, loads are the highest in the tower resulting in the highest shear displacement $u_{sh,1}$ and hence, local structural behaviors such as the response of the sheathing-to-framing connections and axial compression in the studs are interesting to study.

The elasto-plastic sheathing-to-framing connection response in inner layer of OSB (Sh_1) and outer layer of gypsum board (Sh_2) of the front shear wall is shown in Fig. 10*(a)* and *(b)* for vertical load $F_{y,1}$ = 460 kN and the characteristic horizontal load $F_{x,1}$ = 97.5 kN on the module M_1. The blue vectors indicate the elastic response, while the yellow vectors indicate the onset of plasticity with the first yield force P_0 = 200 N, see Fig. 5(*a*) and also [9], within the sheathing-to-framing connections of the shear wall. Similarly, Fig. 10*(c)* and *(d)* show the force vectors during same vertical load and the maximum horizontal load $F_{x,max}$ = 225 kN on the module M_1 for both sheathing panels. It indicates that most of the fasteners have undergone substantial yielding even passing the second yield force P_1 = 400 N, as illustrated by the red vectors. The majority of the red vectors on the wider portion of the shear wall show that the wall has reached its ultimate capacity, thus indicating the maximum load $F_{x,max} = F_{x,1}$ = 225 kN as the shear

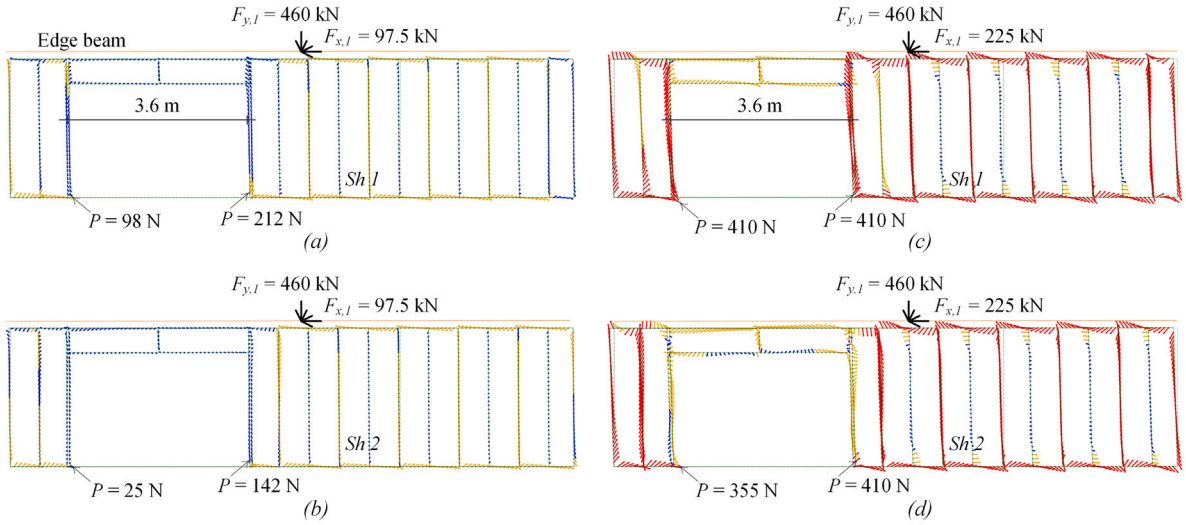


Fig. 10. Illustration of force distribution of the sheathing-to-framing connections with some connection force values in one shear wall of the bottom module M_1 for *(a)* inner layer Sh_1 and (*b*) outer layer Sh_2 of the sheathing panels when $F_{x,1}$ = 97.5 kN, and *(c)* inner layer and (*d*) outer layer when $F_{x,1}$ = 225 kN acting on the module. For interpretation of the references to color in this figure, the reader is referred to the web version of this article.

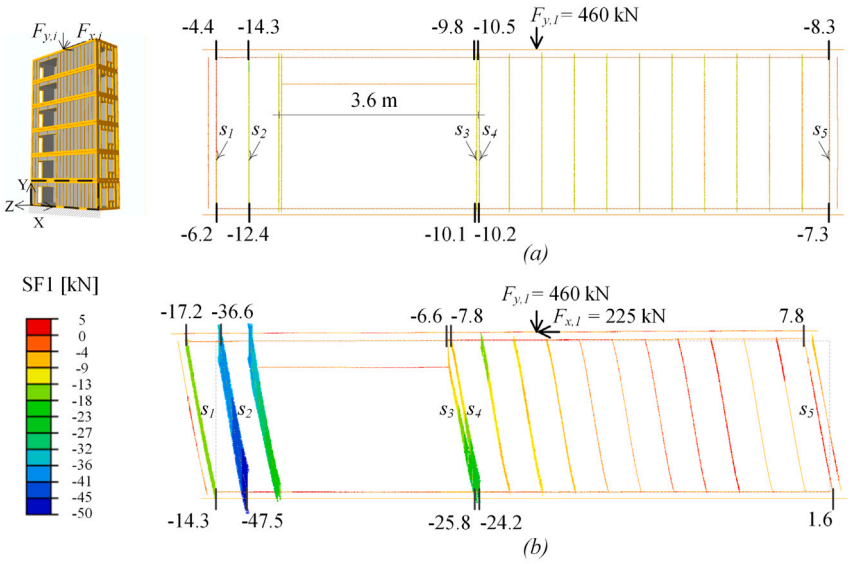


Fig. 11. Illustration of compression forces (in kN) for selected studs $s_1 - s_5$ in the shear wall of the bottom module M_1 due to (*a*) vertical load $F_{y,1}$ and (*b*) combination of vertical $F_{y,1}$ and horizontal load $F_{x,1}$.

capacity of the module. The relatively high forces in the sheathing-to-framing connections suggest that these connections govern the design of the building.

Another critical component to investigate is the performance of the most loaded stud of the shear wall during combined loading. Fig. 11(*a*) shows the axial compression forces in the studs of the front shear wall of the bottom module M_1 under vertical loading, with values given at the top and bottom of some selected studs. Vertical loads are unevenly distributed in the studs under vertical load $F_{y,1}$ = 460 kN in the module, with compressive forces at the bottom $N_{c,2}$ = 12.4 kN on stud s_2 and $N_{c,3}$ = 10.1 kN on stud s_3 which is adjacent to the 3.6 m opening in the shear wall. When horizontal load $F_{x,1}$ = 225 kN is added in the module, the compression force at the bottom of the most loaded stud s_2 increases to $N_{c,2}$ = 47.5 kN, see Fig. 11*(b)*, representing 3.8 times increase compared to vertical loading alone. This high force in s_2 is surprising since the stud is to the left of the relatively big opening but illustrates the role of the top flooring including the edge beam in transferring forces around the opening. The compression forces at the bottom of the double studs s_3 and s_4 have also increased to 25.8 and 24.2 kN respectively, the sum of which is close to the compression force at the bottom of the stud s_2. It should also be recalled that in this model the sheathings are linear elastic so that any cracks that might occur in the physical module is not included in these results which might affect the distribution of forces in the studs. Additionally, uplift effects in the module M_1 change the compressive force of the stud s_5, farthest from the opening from $N_{c,5}$ = 7.3 kN to a tensile force of $N_{t,5}$ = 1.6 kN.

For a 2.4 m C24 stud with a cross-section of 45 × 95 mm, the calculated buckling resistance about the strong axis following EC5 [2] is N_{Rd} = 29.4 kN, which is considerably lower than the simulated axial force during combined loading. This marks the over-utilization of the stud s_2 at the maximum load in the bottom module. Thus, special consideration is necessary while designing the studs beside large openings to prevent buckling failure around the strong axis.

A possible solution is to replace the single stud with multiple studs (two or three), close to openings, similar to the double studs s_3 and s_4 or alternatively to increase the stud cross-section. A study with two

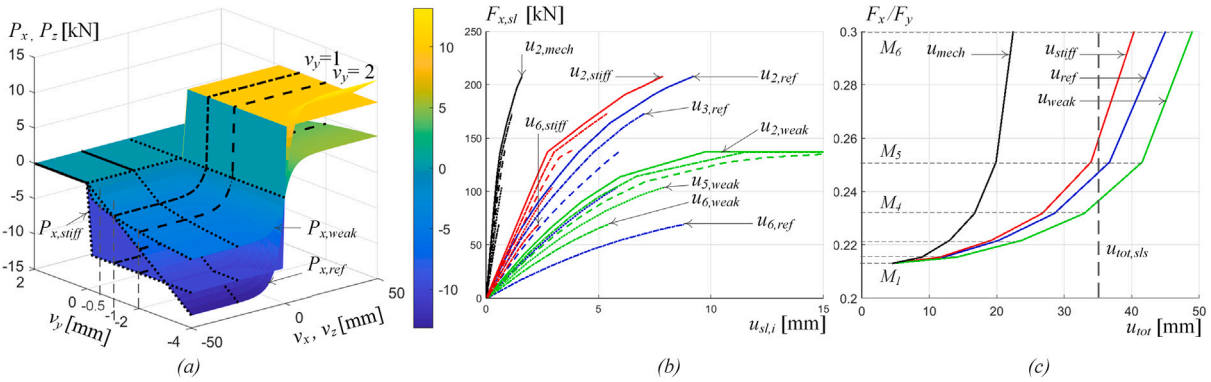


Fig. 12. (*a*) Three surfaces showing coupled relation between P_x-v_x and P_z-v_z with respect to v_y for stiff, reference (ref) and weak model, (*b*) force-slip displacement diagram of each modules for the four models and (*c*) plot for ratio of cumulative forces F_x and F_y versus total horizontal displacement u_{tot} for each story i when $Q_{k,1}$ = 97.5 kN.

studs s_{2a} and s_{2b} replacing the stud s_2 reveals that the compression forces $N_{c,2a}$ = 33 kN and $N_{c,2b}$ = 40 kN respectively, at the maximum horizontal load $F_{x,1}$ = 225 kN. Replacing s_2 with three studs show the forces $N_{c,2a}$ = 30 kN, $N_{c,2b}$ = 35 kN and $N_{c,2c}$ = 38 kN at the same load level. These results indicate that the compression forces in the studs exceed the buckling resistance at horizontal loads $F_{x,1}$ = 156 kN and 188 kN for the double and triple stud configurations respectively, corresponding to an average compression force $N_{c,2avg}$ = 29 kN.

Moreover, the design compressive load perpendicular to the grain for a C24 rail is $F_{c,90,d}$ = 26 kN. The simulations show that this limit is reached in one of the replacement studs in s_2 at horizontal loads of only $F_{x,1}$ = 136 kN for the double stud and 163 kN for the triple stud configuration.

These findings highlight the importance of evaluating local stress concentrations and failure mechanisms in the critical studs of the most heavily loaded bottom module. Particular attention should be given to these studs during the design process to ensure that the racking strength $F_{x,1}$ = 225 kN can be achieved in the bottom module M_1. The results also marks the need for further investigation of compression perpendicular to the grain, which is excluded in the present study.

5. Parametric study

In this section, parametric studies are performed to investigate the effects of variations in inter-modular connector behavior, position of loading, number of stories and types of modules on the towers of the module types previously simulated, thereby providing further insight into the FE model.

5.1. Inter-modular connector behavior

The six-story tower is analyzed by defining three new properties of the inter-modular connection; one stiffer than the reference model previously used and another less stiff (weak) than the reference model. Additionally, a very stiff mechanical (mech) connection is also studied that depicts nail plate with mechanical fasteners. All the alternative behaviors are compared to the results from the simulation of the reference model of module B, see Section 4. The three surfaces that define the coupled mechanical behavior of the connector by relating the horizontal force P_x to the displacements v_x and v_y in the elastomer bricks are illustrated in Fig. 12(*a*) (see also Fig. 4(*d*)). The reference model suggests a linear increase of the $P_{x,ref}$ with respect to the vertical displacement v_y of the elastomer bricks, see [9]. The force P_x = 0 when $v_y \geq 0$, therby preventing in-plane forces from being transmitted through the elastomer bricks during uplift. In addition, the stiff and mechanical definitions of the connection use a horizontal force $P_{x,stiff}$ = ±10 kN when $v_y \leq$ -0.2 mm. The less stiff connection with the asymptotic curve approaching P_{max} is obtained as:

$$P_{x,weak} = P_{x,weak}(v_x, v_y) = \sqrt[3]{v_y} \cdot \left(P_{\max} \cdot \left(1 - e^{-k\cdot|v_x|}\right)\right) \cdot \text{-sign}(v_x) \qquad (19)$$

where $P_{\max}$ = 3.3 kN is the maximum value asymptotically approached when v_x approaches ± 50 mm and v_y = −1 mm, and k = 0.25 represents the rate at which the asymptote is approached. Moreover, the properties of mechanical inter-modular connections are supplemented with the vertical compression stiffness $k_{ey,c,mech}$ = 50 MN/m (10 × $k_{ey,c}$, see Fig. 5(*b*)) and a higher modulus of elasticity of the solid component, E_{mech} = 93 MPa (10 × E_e, see Table 1). The less stiff connection allows only a maximum $P_{x,weak}$ = 5 kN even if the vertical displacement of the connections v_y = −4 mm. Note that for -1 mm < v_y < 0, the reference surface has lower force P_x values than the weak definition. For example, when v_y = −0.5 mm and v_x = 10 mm, $P_{x,ref}$ = 2.3 kN while $P_{x,weak}$ = 3.6 kN. At v_y = −1 mm, $P_{x,ref}$ and $P_{x,weak}$ coincide to P_{max} while $P_{x,stiff}$ = 10 kN. If instead for example v_y = −2 mm, the P_x-v_x relation increases stepwise between the weak, reference and stiff definitions respectively.

Fig. 12(*b*) shows force F_x versus slip displacement $u_{sl,i}$ for each type of connection and for each story. For module M_2, the slip displacement at $F_{x,1}$ = 225 kN for the reference model $u_{2,ref}$ = 9.2 mm while for the stiff model $u_{2,stiff}$ = 7.8 mm implying a difference of only 1.4 mm (15%). On the other hand, the difference was 2.9 mm (49%) for module M_5 at $F_{x,sl,5}$ = 104 kN and peaked to 6.4 mm (74%) for the top module M_6 at load $F_{x,6}$ = 69 kN. The various differences are closely related to the difference between the surfaces at different vertical displacements v_y, see Fig. 12(*a*), caused by different vertical load $F_{y,i}$. It illustrates a noticeable reduction in slip displacement u_{sl} between modules on the upper floors when the stiff definition is used.

Moreover when mechanical connections are used, the slip is significantly reduced to $u_{2,mech}$ = 1.6 mm (83%) for module M_2 and to $u_{6,mech}$ = 0.53 mm (94%) for M_6. The displacement due to rigid body rotation is also reduced by 70% compared to those obtained in the reference model. Thus the mechanical properties significantly affect the lateral response of the multi-story tower. The total horizontal displacement of the top module M_6 for the reference model is u_{ref} = 45.1 mm whereas for the model with mechanical connections the same displacement is u_{mech} = 22.4 mm, which is below the lateral displacement limit criteria in serviceability limit state $u_{tot,sls}$ = 36 mm, when the characteristic load acting on the module M_1 is $Q_{k,1}$ = 97.5 kN, see Fig. 12(*c*). The figure illustrates total horizontal displacement u_{tot} for each story of the models against the ratio of cumulative horizontal F_x and vertical F_y forces acting on the modules.

In contrast, when a weak connection is used, modules M_2, M_3 and M_4 show a considerable increase in $u_{sl,i}$ when $F_{x,sl}$ reaches 137 kN, indicating slip in those modules. In case of module M_6, $u_{6,ref}$ is greater than $u_{6,weak}$ because the vertical displacement of the elastomer brick is v_y = −0.4 mm and for that case $P_{x,ref}$ is less stiff than the $P_{x,weak}$ one, see Fig. 12(*a*).

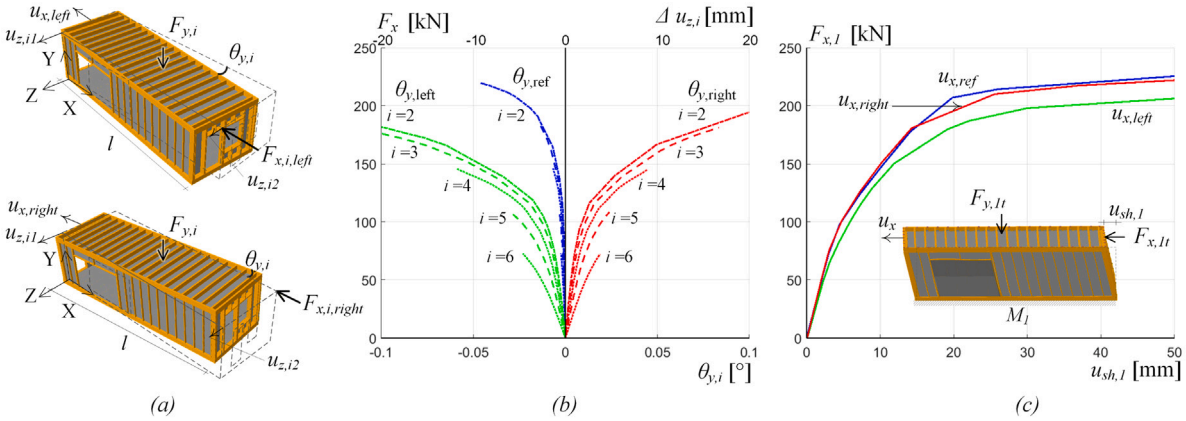


Fig. 13. (*a*) Asymmetric loading illustrating rotational angle $\theta_{y,i}$ and deformed modules during loading on the left $F_{x,i,left}$ and the right $F_{x,i,right}$ side of the module, (*b*) effect in $\theta_{y,i}$ for each story i and (*c*) effect in shear displacement $u_{sh,1}$ for module M_1.

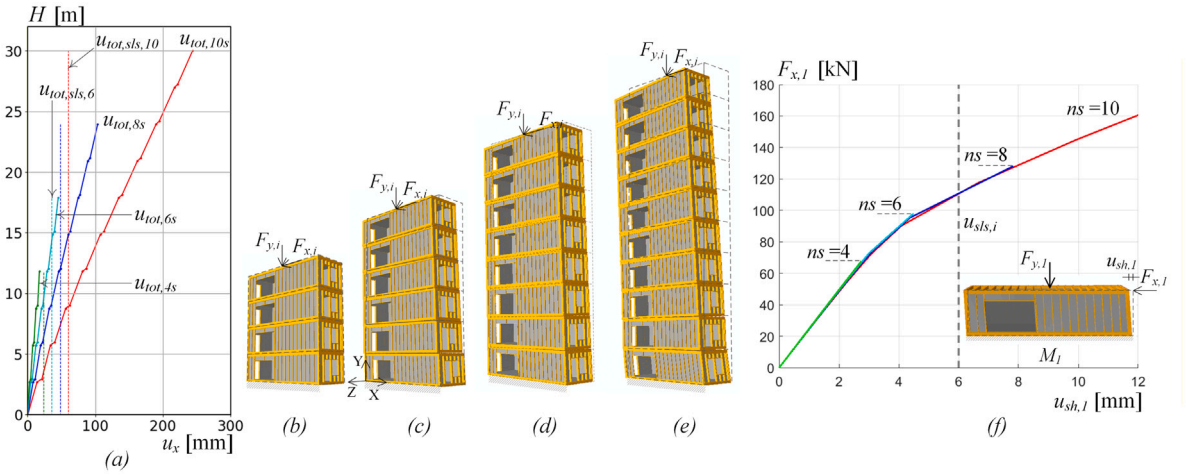


Fig. 14. (a) Total horizontal displacement $u_{tot,ns}$ for different towers studied when the characteristic load on the bottom module is $Q_{k,1,ns}$, deformed models of the tower for stories ns = (*b*) 4, (*c*) 6, (*d*) 8 and (*e*) 10 respectively and (*f*) force displacement curves for shear displacements of the bottom module M_1.

5.2. Position of loading

The six-story tower composed of module B is asymmetric with respect to shear stiffness as one shear wall includes a 3.6 m wide opening while the other is fully sheathed. In addition to the asymmetry caused by the design, the load is applied on the left and right sides of the transverse wall of each module, see Fig. 13(*a*) and the results are compared to those from the reference model which is loaded as discussed in Section 4.1.

The rotational angle around the Y-axis, $\theta_{y,i}$ for each module is calculated by using the horizontal displacement along the Z-axis ($u_{z,i}$) and the length of the modules (l) so that $tan(\theta_{y,i}) = (u_{z,i1} - u_{z,i2})/l = \Delta u_{z,i}/l$. The F_x versus $\theta_{y,i}$ and $\Delta u_{z,i}$ curves for the three load cases are plotted for each module $2 \le i \le 6$, see Fig. 13*(b)*. For the reference model, the rotation $\theta_{y,i,ref}$ is significantly lower than the other two cases and caused purely by the asymmetry in the design. A comparison between the three load cases for module M_2 reveals that at $F_{x,2}$ = 150 kN, $\theta_{y,2,left} \simeq -0.046°$, $\theta_{y,2,ref} \simeq -0.005°$ and $\theta_{y,2,right} \simeq 0.035°$. In other words, at this load for the module M_2, the difference in horizontal Z-displacement between $u_{z,i1}$ and $u_{z,i2}$, see Fig. 13(*a*) is $\Delta u_{z,2,left} = -8.6$ mm and $\Delta u_{z,2,right} = 6.6$ mm, while the reference module has only $\Delta u_{z,2,ref} = -0.9$ mm. This difference shows the substantial influence of the load position. When the module is loaded on the left side where there is an opening, $\theta_{y,i,left}$ is slightly higher than $\theta_{y,i,right}$. This illustrates the significance of the opening position with respect to rotational behavior.

Table 5
Total horizontal displacement $u_{tot,ns}$ and characteristic load $Q_{k,1,ns}$ on the bottom module for ns story tower.

ns	$u_{tot,ns}$ [mm]	$u_{tot,sls,ns}$ [mm]	$Q_{k,1,ns}$ [kN]	$u_{sh,1,ns}$ [mm]	$u_{sh,1,ns}/u_{sls,i}$ [–]
4	18	24	67.5	2.8	0.46
6	45	36	97.5	4.5	0.75
8	104	48	127.5	7.8	1.30
10	244	60	157.5	11.6	1.93

Furthermore, the impact of the load position is studied in the shear displacement of the bottom module M_1. The horizontal force $F_{x,1}$ is plotted against the shear displacement $u_{sh,1}$ taken in the middle of the top part of the transverse wall (u_x), see Fig. 13*(c)* for three loading sequences. The stiffness for the reference model and the model loaded on the right with no opening ($u_{x,right}$) are similar with a slight reduction in racking strength for the latter one. However, when it is loaded on the left side with an opening ($u_{x,left}$), not only is the stiffness reduced, but the racking strength of the module is also negatively affected by around 8.5%.

5.3. Number of stories

The total horizontal displacement $u_{tot,ns}$ is analyzed for module B towers with four, eight and ten number of stories (ns) and compared

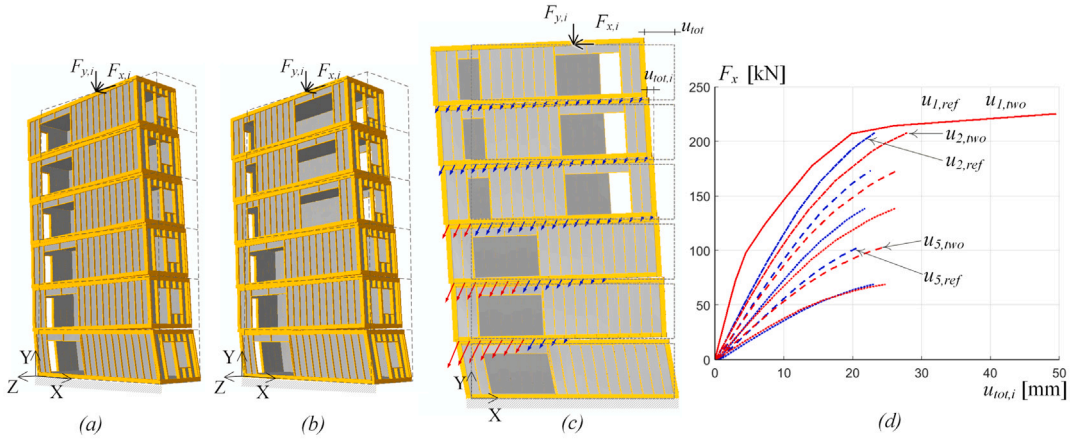


Fig. 15. Illustration of (*a*) deformed shape of the reference model, (*b*) deformed shape of tower composed of two modules type B and C, (*c*) forces in elastomer bricks and (*d*) force–displacement diagram showing horizontal force F_x and story drift $u_{tot,i}$.

with the reference model tower with $ns = 6$. The displacement $u_{tot,ns}$ is illustrated along with the deformed shapes for ns story towers when the force $Q_{k,1,ns}$, acting on the bottom module M_1 caused by the horizontal characteristic load, is reached, see Fig. 14 and Table 5. For the six-story reference model, $u_{tot,6s} = 45$ mm, whereas the limit value $u_{tot,sls,6} = 36$ mm. Moreover, the eight-story tower has $u_{tot,8s} = 104$ mm while $u_{tot,sls,8} = 48$ mm and the ten-story tower has $u_{tot,10s} = 244$ mm while $u_{tot,sls,10} = 60$ mm. Hence, the total horizontal displacement has surpassed the limit for models equal to or taller than the six-story tower. However, for the four-story model, $u_{tot,4s} = 18$ mm which is within the limit $u_{tot,sls,4} = 24$ mm.

It is also of interest to check the limit value for shear displacement $u_{sh,1}$ of the bottom module M_1 during the horizontal loading, and therefore the force displacement diagram is plotted, see Fig. 14*(f)*. Table 5 and the figure illustrate that the four and the six-story tower ($u_{sh,1,4} = 2.8$ mm and $u_{sh,1,6} = 4.5$ mm) are within the serviceability limit while the eight and the ten-story tower ($u_{sh,1,8} = 7.8$ mm and $u_{sh,1,10} = 11.6$ mm) are beyond the limit value $u_{sls,i} = 6$ mm. From previous results, see Fig. 6(*c*), it is clear that the linear approximation introduced for substructure stories $i \geq 2$ results in too small shear displacements. It should therefore, be noted that the displacements in Fig. 14 are underestimated or on the low side.

5.4. Types of modules

Substructure analysis is performed for two module types, which are subsequently combined to develop the adaptive model of a six-story tower. The model consists of a standard FE model for the bottom module, accompanied by two substructure models of module B and three substructure models of module C, see Section 2.1, stacked above. In this configuration, the position of the elastomer bricks differ from those in the previous tower model because they are aligned with the stud locations and the position of the openings are different from the previous model. The deformed shape of the reference model composed of module B, Section 4.1, and the model with two types of module are illustrated in Fig. 15*(a)* and *(b)* respectively, when the horizontal force $F_{x,1} = 225$ kN. In general, the deformed shape of the modules for both tower models is similar, indicating a limited effect of the two combined configurations.

Fig. 15*(c)* illustrates internal forces acting on elastomer bricks of the model with two types of module and Fig. 15*(d)* shows the force–displacement diagram for horizontal force F_x and story drift $u_{tot,i}$ for the reference model (ref) and the model with two module types (two). The curve for module M_1 is the same in both cases; however, for modules M_2–M_6, the story drift for the model with two types of modules $u_{i,two}$ $\simeq$ 16% higher than those observed in the reference model. This discrepancy is due to increased slip displacement $u_{sl,i}$ of modules M_2–M_6 ($\simeq$92%) and increased shear displacement $u_{sh,i}$ of module C at M_4–M_6 ($\simeq$8%). The increased slip is the result of reduction of vertical force F_y distributed in each elastomer bricks (due to increased number of bricks) eventually reducing v_y and P_x, see $P_{x,ref}$ surface in Fig. 12(*a*), so that slip occur at lower load compared to the reference model. Similarly the increased shear displacement of module C is due to supplemented door openings.

6. Conclusions

The study presented the adaptive modeling strategy for light-frame timber modular towers that balances computational efficiency and numerical accuracy. The approach integrates a detailed nonlinear standard FE-formulation for the critical bottom module subjected to the highest loads with a linear substructure formulation for the remaining modules stacked in a tower configuration. This method enables efficient simulation of large modular assemblies while accurately capturing the nonlinear (plastic) behavior of key structural regions, such as inter-modular connections and critical shear walls. The numerical verification of the substructures and resource efficient modeling using repeated substructure assemblies were discussed. The adaptive model allows extraction of both racking stiffness and racking strength of both the modules and the multi-story tower, and total horizontal displacements were subsequently compared against the recommended serviceability limit state design displacements.

The major findings of the study are summarized as follows.

1. The substructure model of the single module A reproduces the shear displacement u_{sh} accurately up to a horizontal load $F_x = 50$ kN, showing a strong agreement with the standard FE model and confirming its reliability up to that force level with a computational efficiency by reducing the CPU time by 68% compared to the standard FE-formulation.
2. Comparison of the components of the story drift $u_{tot,i}$ of the six-story tower composed of module A reveals that the displacement due to rigid body rotation $u_{br,i}$ contributes more (about 40.3% in total horizontal displacement u_{tot}) than the shear $u_{sh,i}$ and slip displacement $u_{sl,i}$. While $u_{sh,i}$ decreases from bottom to the top, $u_{br,i}$ increases. The slip displacement $u_{sl,i}$, caused by the inter-modular connections, remains uniform across stories contributing about 25% in u_{tot}.
3. For the tower composed of a typical asymmetric module B, the racking stiffness and racking strength of the bottom module are determined as $K_1 = 23$ kN/mm and $F_{x,1} = 225$ kN respectively.

The combined effects of the accumulated vertical and horizontal loading concentrate at the bottom module, which justifies the standard FE-formulation for this module.

4. Comparison with serviceability limits indicates that the story drift $u_{tot,1}$ of only the bottom module M_1 of the six-story tower composed of module B satisfies the limit value $u_{sls,i}$ = 6 mm under the influence of the vertical load $F_{y,1}$ = 460 kN and the horizontal characteristic load $F_{x,1}$ = 97.5 kN on the module. It should be noted that $u_{tot,i}$ can be easily reduced by designing inter-modular connections in a stiffer way by using metal plates and mechanical fastener. Such connections can reduce $u_{br,i}$ and $u_{sl,i}$ by 69% and 86% respectively but with the same shear displacement u_{sh}, resulting in total horizontal displacement u_{tot} below the serviceability limit state criteria.
5. The standard FE model enables an easily assessable detailed evaluation of local responses such as sheathing-to-framing connections behavior and the compressive forces in the studs of the shear wall.
6. Parametric studies demonstrates a clear influence of inter-modular connections, asymmetric loading, number of stories and types of modules on the overall lateral response of the towers.

The proposed modeling approach offers a resource efficient framework for analyzing multi-story light-frame timber modular towers. It ensures a sufficiently accurate prediction of both global and local behavior, supporting a performance based design and optimization of modular timber structures.

Future works

The development of the adaptive model of multi-story towers could be expanded to accommodate two or multiple towers placed side by side. The analysis could also be expanded to modules ranging from two to multiple types of modules to design an entire modular building and to achieve efficient yet effective simulation of the building. Moreover, there are possibilities to improve the adaptive model by including inter-modular connections in horizontal direction, effects of compression perpendicular to the grain in bottom rails and experimental verification for horizontal force P_x vs vertical displacement v_y relation for different types of elastomer bricks. The system effect of the modular building including intra-modular and inter-modular (vertical and horizontal) can be further investigated to fully apprehend the building made up of prefabricated modules.

CRediT authorship contribution statement

Rajan Maharjan: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Johan Vessby:** Writing – review & editing, Resources, Methodology, Conceptualization. **Le Kuai:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis. **Sigurdur Ormarsson:** Writing – review & editing, Resources.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The research has received generous financial support from multiple wood construction companies in Sweden, Formas (2019-01431), Smart Built Environment (2020-00224) and Shanghai Pujiang Program (25PJA051) . The authors express their sincere gratitude for this invaluable assistance.

Data availability

Data will be made available on request.

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